Extended circular-arc Śrī Yantras

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Abstract

Carrying the outermost side of the Śrī Yantra on each hand out until it closes into a triangle of its own gives a figure of eleven triangles, six pointing down and five up. The two new triangles have as legs the curve through the three outer corners of the opposite family, and between them and the circumscribed circle they cut a further ring of cells. We prove two results about these extended Śrī Yantras. First, none has straight sides: no laminar straight Śrī Yantra has the outer corners of both families collinear with a laminar extension. A projective symmetry of the straight figure reduces the statement to one about thirteen points on the projective line, eight equations and twenty-five inequalities between products of cross-ratios, and this statement is decided by a computer-assisted proof. The configuration space is covered by 58,786 cones, one for each triangulation of a 13-gon, on each of which every condition is a linear form in logarithmic coordinates plus corrections that vanish exponentially towards the vertex of the cone. Each cone is closed by interval arithmetic and by linear-programming certificates verified in exact rational arithmetic, except near one vertex of the associahedron, where the equations and the inequalities approach one another and an asymptotic analysis in adapted coordinates decides. Second, with circular arcs the extension exists: we give two laminar figures, whose new rings have ten and fourteen cells, by straightedge and compass constructions in which every incidence holds by construction, with 22 and 20 free parameters.

1 Introduction

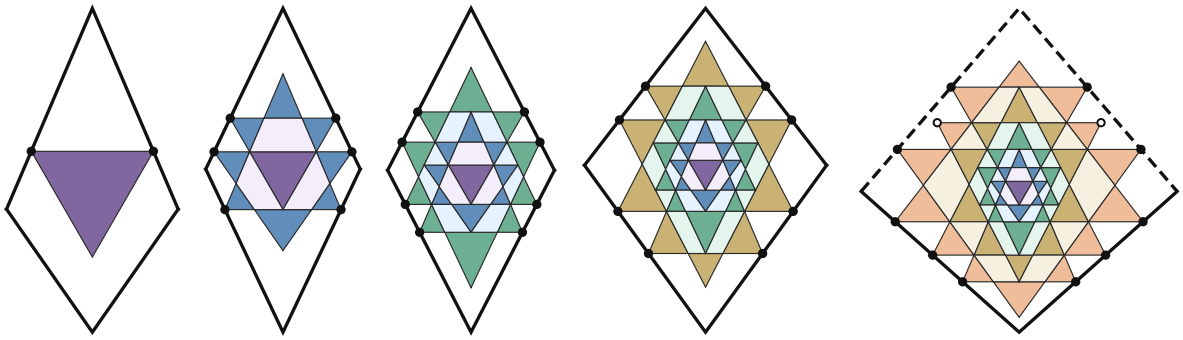
The Śrī Yantra is the union of nine triangles, five pointing down and four up, symmetric about a common axis, whose sides are required to pass through a large number of common points. Chiodo [2] gave a straightedge and compass construction of the straight figure and identified its essential step as a circle–line–point problem of Apollonius. In [3] the twenty-seven sides were replaced by circular arcs, every classical incidence was kept, and the fourteen outer corners of the figure were placed on the circumscribed circle; the result is an 18-dimensional family reached by a straightedge and compass construction in which every coincidence holds identically. We follow the notation of [2, 3] throughout.

The traditional diagram continues outwards. Around the nine triangles lie the enclosures of eight and sixteen lotus petals, three circles and the square *bhūpura*; inside the triangles the outermost enclosure is the *caturdaśāra*, the ring of fourteen triangles whose outer vertices are the fourteen outer corners. The present paper concerns the natural question of whether the triangular structure itself can be continued by one more ring.

The rings of the classical figure are nested in a way that indicates how such a ring is to be added (Figure 1). For an upward triangle U and a downward triangle D whose apex lies below that of U, we call the quadrilateral bounded by their legs, from the two apexes to the

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(a) U_3, D_5 (b) U_4, D_4 (c) U_2, D_2 (d) U_1, D_1 (e) U_0, D_0

Figure 1: A laminar straight Śrī Yantra of classical type in which the outer corners of U_1, U_2, U_3 are collinear, built from the centre outward. (a) The central triangle, with its base corners (dots) on the legs of U_3, in the kite of U_3 and D_5. (b)–(d) The rings of eight, ten and ten triangles, each with its top and bottom vertices at the apexes of the previous kite and its other outer vertices (dots) on the legs of the next: the kites of U_4 and D_4, of U_2 and D_2, and of U_1 and D_1. (e) The ring of fourteen. Its outer vertices off the axis are the outer corners of D_1, D_2, D_3 and of U_1, U_2, U_3, and a kite through them is the extension. Here the corners of U_1, U_2, U_3 lie on the lower legs of the kite (solid); the upper legs, drawn through the corners of D_1 and D_3 (dashed), miss the corner of D_2 (open dots). Each ring is shaded in its own tone, lighter outward, and the cells between consecutive rings in a pale tint of the inner one. Each panel is scaled to its kite.

two points where a leg of U crosses a leg of D, the *kite* of U and D. The central triangle lies in the kite of U_3 and D_5, with its two base corners on the legs of U_3. Each of the rings of eight, ten and ten triangles around it has its top and bottom vertices at the apexes of one kite and its other outer vertices on the legs of the next, the kites being, from the inside, those of U_3 and D_5, U_4 and D_4, U_2 and D_2, and U_1 and D_1; the ring of fourteen has its top and bottom vertices at the apexes of the last. On each side of the axis these are twelve incidences of a vertex with a leg, and they are exactly Chiodo's twelve concurrences (Section 3). Read from the centre outward, the figure is built ring by ring: the lines already drawn are carried on beyond their kite, they cut out the next ring, and the outer vertices of that ring fix the next kite.

In this reading the ring of fourteen is the last ring built, and a further ring needs a fifth kite, whose legs carry the outer corners of the ring of fourteen. In the traditional drawing the outer corners of the three largest downward triangles lie close to a common line, and so do those of the three largest upward triangles; in the yantra of Figure 1 the upward triple is collinear and the downward triple is not. If each triple lies exactly on a common side, that side and its mirror image close into a new triangle: an upward triangle U_0 whose legs pass through the outer corners of D_1, D_2, D_3, and a downward triangle D_0 whose legs pass through those of U_1, U_2, U_3. Their kite is the fifth kite, and the figure has eleven triangles, six down and five up. Carried on beyond the fifth kite, as at every earlier step, the sides of U_0 and D_0 cut out a new ring. With straight sides it consists of the tips of U_0 and D_0 at their apexes and the four triangles at the corners of their bases (Figure 2), the six cells of the pattern without walls in Section 8; with the circular arcs of Part II it is cut out of the disc between the yantra and the circumscribed circle, the rim. We call such a figure an *extended* Śrī Yantra.

The figure is asked to keep the character of the original in one respect that is not an incidence. Within each family the triangles of the Śrī Yantra do not cross: no leg of a downward triangle meets the leg of another downward triangle, and likewise upwards. We call this property *laminarity* (Definition 2.1); it is what gives the figure its characteristic layered appearance, and it is the property any extension has to preserve.

What is proved

Theorem A. There is no straight extended Śrī Yantra of classical type. That is, in no straight Śrī Yantra whose nine triangles are laminar can the outer corners of D_1, D_2, D_3 and those of U_1, U_2, U_3 both be collinear in such a way that the two triangles they span contain their families and the eleven triangles are laminar.

Figure 2 shows how close the straight figure comes. Its eleven triangles are laminar and both triples of outer corners are collinear, so that the fifth kite exists, and every incidence of the Śrī Yantra holds except one: the corner of U_4, an outer vertex of the inner ring of ten, misses the leg of D_2, a side of the third kite. Such figures form a family of dimension four, and Theorem A says that along it the miss cannot be brought to zero without breaking laminarity. The two are in tension well before that point. Laminarity is a finite list of inequalities (Section 3), and we measure each by the ratio of its two sides minus one. In our computations the smallest of these relative clearances can be raised to about 16% when the miss is 10% of the half-width of U_4, and only to about 4% when the miss is 4%; the clearances that limit it are those between the leg of D_4 and the corner of D_5 and between the corner of U_1 and the side of U_0. The figure shown trades closeness for legibility: its miss is 15% and its smallest clearance about 20%.

The proof has two parts of different character. The first is an exact reduction (Sections 3–4). A straight Śrī Yantra is determined, up to the horizontal scale, by the eleven heights on the axis at which its apexes and bases lie, subject to four quartic equations; its extension adds two collinearities and the apexes z_D, z_U of the new triangles. Every one of these conditions, and every laminarity condition, is an identity or an inequality between products of differences of the thirteen heights in which each height occurs as often in the numerator as in the denominator. Such products are invariant under all projective transformations of the axis, and the problem therefore lives on the space $\mathcal{M}$ of thirteen cyclically ordered points on the projective line modulo $\mathrm{PGL}_2(\mathbb{R})$, which has dimension ten. In $\mathcal{M}$ the eight equations cut out a surface $\mathcal{V}$ and the inequalities a closed set $\mathcal{L}$, and Theorem A follows from

$$\mathcal{V} \cap \mathcal{L} = \varnothing \qquad \text{(Theorem 4.2).}$$

The second part proves this statement, and it is computer-assisted (Sections 5–7). Its organising fact is that $\mathcal{M}$ is the interior of an associahedron, and that near each of its 58,786 vertices, which correspond to the triangulations of a 13-gon, there are coordinates in which the logarithm of every distance between two of the thirteen points is a linear form plus a correction that vanishes exponentially fast towards the vertex. We use the logarithms of the gaps between consecutive points; the vertices then correspond to the Cartesian trees of the gap sequence, and each cone is a product of half-lines. On each cone the eight equations and the twenty-five inequalities become linear conditions perturbed by small, explicitly bounded, corrections, which makes a branch and bound over half-infinite boxes possible: a box is closed when interval arithmetic, or a linear program whose infeasibility is certified by Farkas multipliers checked in exact rational arithmetic, shows that it contains no point of $\mathcal{V} \cap \mathcal{L}$. This closes every cone except near one vertex T^*. There $\mathcal{V}$ and $\mathcal{L}$ approach each other: the laminar points that satisfy seven of the eight equations accumulate at T^*, and the residual of the eighth tends to zero with them. Near T^* we use coordinates adapted to the vertex, in which a combination Φ of the equations has the explicit leading term $t_3t_4 + t_5t_6 + t_9t_{10}$ (Proposition 6.1), and a certificate in those coordinates closes the neighbourhood (Section 6).

With circular arcs the extension exists. The ring of cells that U_0 and D_0 cut out has, on each of its two sites, room for up to four triangles, separated by three walls; a wall stands when the triangle behind it is carried out to the rim, and there are $2^3 \times 2^3 = 64$ patterns of walls (Section 8.2). For two of them, with ten and fourteen cells in the new ring, we give constructions of the kind [3] gives for the nine-triangle figure.

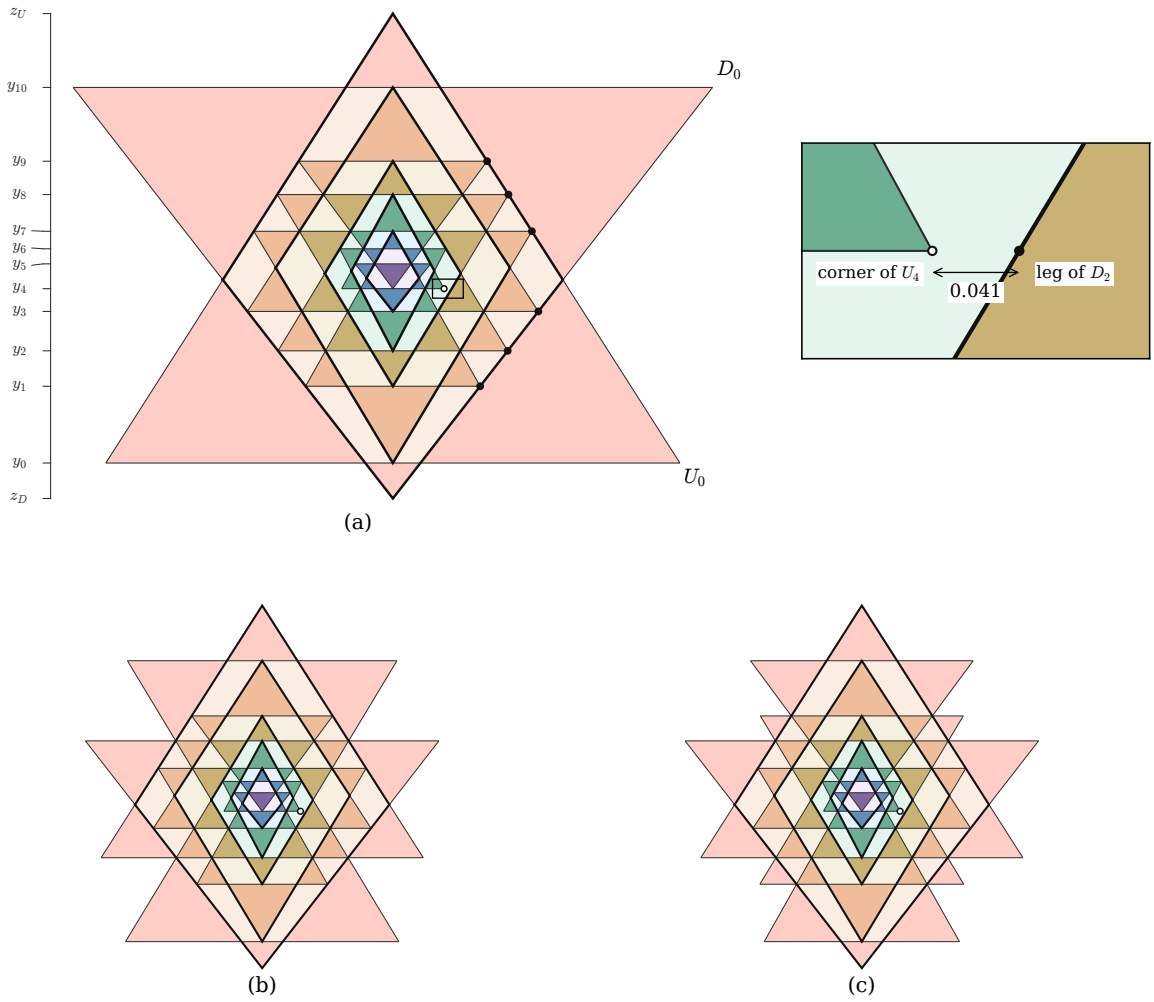


Figure 2: A laminar straight Śrī Yantra with both triples of outer corners collinear (dots), extended by U_0 and D_0, shaded as in Figure 1 and with its five kites drawn heavy. (a) The figure as constructed. Its new ring consists of the tips of U_0 and D_0 at their apexes and the four triangles at the corners of their bases, the six cells of the wall pattern 00 of Section 8. Of the twelve concurrences of the Śrī Yantra eleven hold. The twelfth, ((iii); t_2, t_6, t_6) in Chiodo's notation, asks the leg of D_2, a side of the kite of U_2 and D_2, to pass through the base corner of U_4 (open dot), and misses it by 15% of the half-width of U_4 (right, enlarged from the box; the number marked is the miss in units in which the yantra has height 2). Within the family of such figures this one was chosen for even rings and wide clearances: in each of the rings of eight, ten, ten and fourteen triangles, and among the fourteen cells between the yantra and the fifth kite, the largest has at most 5.5 times the area of the smallest, and every laminarity inequality holds with a relative margin of about 20% or more. The apexes of U_0 and D_0 have been placed at $z_U - y_{10} = y_{10} - y_9$ and $y_0 - z_D = y_2 - y_1$. The horizontal scale is free and has been chosen for legibility; the levels $y_0, \ldots, y_{10}$ and the apexes z_D, z_U of D_0 and U_0 are marked on the left. (b), (c) The same figure with triangles carried out beyond the fifth kite as in Section 8, with horizontal bases in place of the arcs, in the positions of the wall patterns 22 and 36 of Figure 3: the middle corner on each hand (b) and the two outermost (c). On the straight figure these are the corners of D_2 and U_2, and of D_3, D_2 and U_3, U_2, since the straight and the circular-arc figures order their downward bases differently (Section 2). The new rings have ten and fourteen cells.

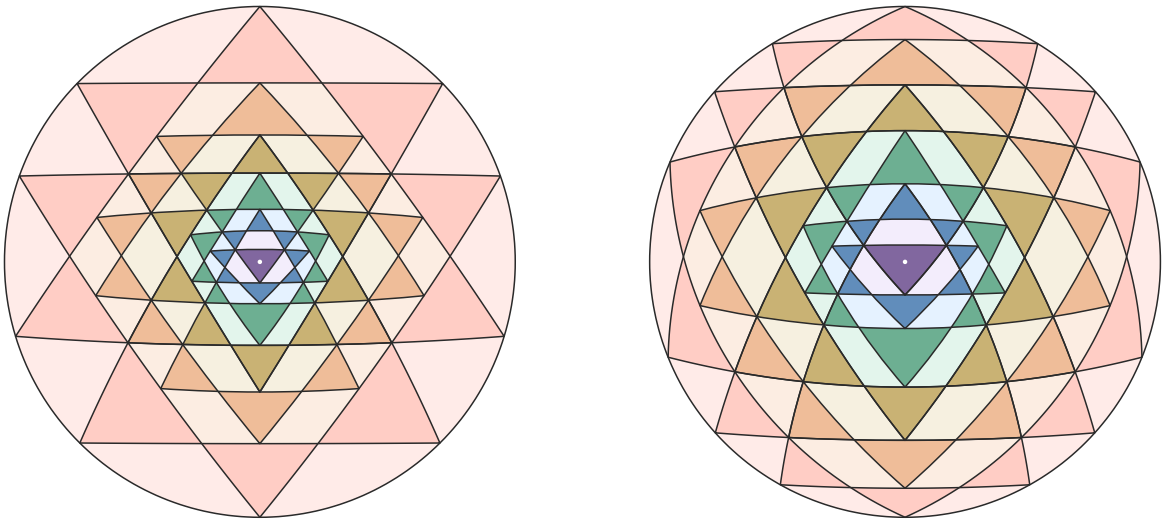


Figure 3: The two constructed extended Śrī Yantras of Theorem B, with ten (left) and fourteen (right) cells in the ring between the yantra and the rim, shaded as in Figure 1; the ground between the new ring and the rim takes the pale tint of the new ring. Every node and every arc is produced by straightedge and compass from the placed corners; the outer nodes of the ring are on the rim by construction. Figure 2(b), (c) shows the straight figure carried out in the same positions.

Theorem B. The circular-arc extended Śrī Yantras with the wall patterns $\{D_3\} \times \{U_2\}$ and $\{D_1, D_3\} \times \{U_2, U_3\}$ are reached by straightedge and compass constructions from 13 placed corners, which carry 22 and 20 free real parameters, and the two poles of the rim:

(i) after the placements the constructions take 52 and 54 steps, each of which is a circle through two or three points or the meeting of a circle with a circle, the axis or the rim;

(ii) every incidence of the figure, including every outer node of its new ring lying on the rim, holds by construction, each realised by exactly one step;

(iii) at the corners of Table 4 the eleven triangles are laminar.

Figure 3 shows the two figures of Theorem B. Section 9 gives the constructions; the accompanying viewer draws both and the GeoGebra files carry them as command lists.

Organisation. Section 2 fixes the objects. Part I (Sections 3–7) proves Theorem A. Part II (Sections 8–9) treats the circular-arc figures and proves Theorem B. Section 10 describes the software and the data that accompany the paper.

2 The extended figure

2.1 Notation

We use the labels of the accompanying software: $D_1, \dots, D_5$ for the downward and $U_1, \dots, U_4$ for the upward triangles. Each triangle is isosceles with its apex on the axis $x = 0$ and its base perpendicular to the axis, or, for arcs, symmetric about it; its two remaining sides are its *legs*. The base corner on the right is the triangle's *corner*. Chiodo's numbering $t_1, \dots, t_9$ orders the triangles by the height of their bases from the top and therefore depends on the figure; on the straight figures considered in Part I it reads

$$t_1 = D_3,\ t_2 = D_2,\ t_3 = D_1,\ t_4 = D_4,\ t_5 = D_5,\ t_6 = U_4,\ t_7 = U_1,\ t_8 = U_2,\ t_9 = U_3,$$

and on the circular-arc figures of Part II the first three are instead $t_1 = D_1$, $t_2 = D_3$, $t_3 = D_2$ [3]. Chiodo's conditions are recalled in Section 3. The upward triangles form one *family* and the downward triangles the other; the new triangles U_0 and D_0 belong to the upward and the downward family respectively.

2.2 Laminarity

Definition 2.1. A figure of triangles divided into two families is *laminar* if no leg of a triangle meets a leg of another triangle of the same family.

The incidences of the Śrī Yantra never ask two legs of one family to meet: each of Chiodo's triple points lies on one downward leg, one upward leg and one base. A base, on the other hand, routinely carries a node on a leg of its own family, and Definition 2.1 accordingly concerns legs only.

2.3 The extension

Let Y be a Śrī Yantra and suppose that the corners of D_1, D_2, D_3 lie on one side ℓ_U, a line or an arc, meeting the axis above the yantra, and the corners of U_1, U_2, U_3 on one side ℓ_D meeting the axis below it. The *extension* of Y consists of the eleven triangles of Y together with

- the upward triangle U_0 with apex $\ell_U \cap \{x = 0\}$, legs ℓ_U and its mirror image, and base through the lowest apex of Y, the apex of D_1;
- the downward triangle D_0 with apex $\ell_D \cap \{x = 0\}$, legs ℓ_D and its mirror image, and base through the highest apex of Y, the apex of U_1.

Thus each new triangle stands on the tip of the opposite family. In the straight case the bases are horizontal; in the circular case they are the arcs described in Section 8. We ask in addition that U_0 contain the four upward triangles of Y and D_0 the five downward ones: the new triangles are the outermost of their families.

Definition 2.2. An *extended Śrī Yantra* is a Śrī Yantra together with its extension, in which U_0 and D_0 are triangles containing the triangles of their families, and the eleven triangles are laminar.

I. Straight sides

3 The straight figure as a ladder of heights

3.1 Chiodo's conditions and the levels

A straight Śrī Yantra is a figure of nine isosceles triangles with horizontal bases and apexes on the axis, satisfying Chiodo's conditions

(ii) $\operatorname{apex}(t') = \operatorname{bp}(t'')$, the midpoint of the base, for the seven pairs

$$(t', t'') \in \{(t_8, t_1),\ (t_6, t_2),\ (t_9, t_3),\ (t_1, t_6),\ (t_5, t_7),\ (t_4, t_8),\ (t_2, t_9)\};$$

(iii) for each of the twelve triples (t', t'', t''') in

$$(t_1, t_2, t_7),\ (t_2, t_3, t_7),\ (t_1, t_3, t_8),\ (t_1, t_4, t_6),\ (t_1, t_5, t_9),\ (t_4, t_6, t_9),$$
$$(t_2, t_7, t_9),\ (t_3, t_7, t_8),\ (t_3, t_8, t_9),\ (t_4, t_4, t_8),\ (t_5, t_5, t_6),\ (t_2, t_6, t_6),$$

a leg of the downward triangle t', the base of t'' and a leg of the upward triangle t''' have a point in common.

Chiodo's condition (i), that t_3 and t_7 be inscribed in a common circle, plays no role here and is not imposed. Condition (ii) identifies seven of the eighteen apex and base heights with others, and the nine triangles therefore have their apexes and bases on eleven heights. We call a straight Śrī Yantra of *classical type* if these levels $y_0 < y_1 < \cdots < y_{10}$ carry the apexes and bases as in the classical figure, listed in Table 1.

	D_1	D_2	D_3	D_4	D_5	U_1	U_2	U_3	U_4
apex	y_0	y_1	y_4	y_2	y_3	y_{10}	y_9	y_7	y_8
base	y_7	y_8	y_9	y_6	y_5	y_3	y_2	y_1	y_4

Table 1: The levels of the apexes and bases of a straight Śrī Yantra of classical type. The seven coincidences of Chiodo's condition (ii) are the seven levels that carry an apex and a base.

We write $y_{ij} = y_j - y_i > 0$ for $i < j$. For a triangle T with apex level a_T, base level b_T and base half-width W_T, the half-width of T at height y is

$$w_T(y) = W_T \, \frac{y - a_T}{b_T - a_T},$$

the abscissa of its right leg at that height. Each triple point of condition (iii) lies on the base of t'', at height $y = b_{t''}$, and it is the statement $w_{t'}(y) = w_{t'''}(y)$: the right legs of t' and t''' meet at that height. A horizontal stretch $x \mapsto \lambda x$ preserves every condition; we fix it by giving D_1 a right angle at its apex, $W_{D_1} = y_{07}$.

Lemma 3.1. *Let $y_0 < \cdots < y_{10}$. Eight of the twelve conditions (iii), namely those for the triples (t_1,t_2,t_7), (t_1,t_4,t_6), (t_1,t_5,t_9), (t_4,t_6,t_9), (t_2,t_7,t_9), (t_3,t_7,t_8), (t_3,t_8,t_9) and (t_5,t_5,t_6), hold if and only if the half-widths are*

$$\begin{aligned}
&W_{D_1} = y_{07}, && W_{D_2} = \frac{y_{18}\, y_{02}\, y_{37}}{y_{27}\, y_{13}}, && W_{D_3} = \frac{y_{49}\, y_{02}\, y_{57}}{y_{27}\, y_{45}},\\
&W_{D_4} = \frac{y_{26}\, y_{02}\, y_{47}}{y_{27}\, y_{24}}, && W_{D_5} = \frac{y_{02}\, y_{57}\, y_{46}\, y_{58}}{y_{27}\, y_{45}\, y_{68}}, && W_{U_1} = \frac{y_{3,10}\, y_{02}\, y_{57}\, y_{48}}{y_{27}\, y_{45}\, y_{8,10}},\\
&W_{U_2} = \frac{y_{29}\, y_{03}}{y_{39}}, && W_{U_3} = \frac{y_{17}\, y_{02}}{y_{27}}, && W_{U_4} = \frac{y_{48}\, y_{02}\, y_{57}\, y_{46}}{y_{27}\, y_{45}\, y_{68}}.
\end{aligned} \tag{1}$$

Given (1)*, the remaining four conditions (iii) are equivalent to the quartic equations*

$$\begin{aligned}
(t_2,t_6,t_6):&\quad y_{14}\, y_{37}\, y_{45}\, y_{68} = y_{13}\, y_{46}\, y_{48}\, y_{57}, && (E_1)\\
(t_4,t_4,t_8):&\quad y_{03}\, y_{24}\, y_{27}\, y_{69} = y_{02}\, y_{26}\, y_{39}\, y_{47}, && (E_2)\\
(t_1,t_3,t_8):&\quad y_{03}\, y_{27}\, y_{45}\, y_{79} = y_{02}\, y_{39}\, y_{47}\, y_{57}, && (E_3)\\
(t_2,t_3,t_7):&\quad y_{13}\, y_{48}\, y_{57}\, y_{7,10} = y_{17}\, y_{37}\, y_{45}\, y_{8,10}. && (E_4)
\end{aligned}$$

Proof. Each condition (iii) for a triple (t',t'',t''') is the linear relation $W_{t'}\,(b_{t''}-a_{t'})/(b_{t'}-a_{t'}) = W_{t'''}\,(b_{t''}-a_{t'''})/(b_{t'''}-a_{t'''})$ between two half-widths, with coefficients that are ratios of level differences (Table 1). The eight triples listed join the nine triangles in a tree,

$$D_3\text{–}U_1,\quad D_3\text{–}U_4,\quad D_3\text{–}U_3,\quad U_3\text{–}D_4,\quad U_3\text{–}D_2,\quad U_3\text{–}D_1,\quad D_1\text{–}U_2,\quad U_4\text{–}D_5,$$

so with W_{D_1} fixed they determine the other eight half-widths uniquely, and solving along the tree gives (1). Substituting (1) into the four remaining relations and clearing the positive denominators gives (E_1)–(E_4); for instance (t_2,t_6,t_6) asks the leg of D_2 to pass through the corner of U_4, $w_{D_2}(y_4) = W_{U_4}$, and $w_{D_2}(y_4)/W_{U_4}$ equals the quotient of the two sides of (E_1). All twelve identities have been verified in exact rational arithmetic at random rational levels (`check/straight.py`). □

A straight Śrī Yantra of classical type is therefore the same as a ladder $y_0 < \cdots < y_{10}$ satisfying (E_1)–(E_4), determined up to the affine maps of the axis and the horizontal stretch. Modulo the affine maps the ladders form a nine-dimensional space, and the family of straight Śrī Yantras has dimension five, Chiodo's condition (i) then cutting it to four.

3.2 The extension

The corners of D_1, D_2, D_3 are the points (W_{D_1}, y_7), (W_{D_2}, y_8), (W_{D_3}, y_9), and those of U_3, U_2, U_1 are (W_{U_3}, y_1), (W_{U_2}, y_2), (W_{U_1}, y_3). Each triple is collinear exactly when its middle point lies on the segment joining the other two:

$$W_{D_2}\, y_{79} = W_{D_1}\, y_{89} + W_{D_3}\, y_{78}, \qquad (C_D) \tag{2}$$

$$W_{U_2}\, y_{13} = W_{U_3}\, y_{23} + W_{U_1}\, y_{12}. \qquad (C_U) \tag{3}$$

The line ℓ_U through the corners of D_1 and D_3 meets the axis at the height z_U with

$$W_{D_1}(z_U - y_9) = W_{D_3}(z_U - y_7), \qquad (A_U)$$

and the line ℓ_D through the corners of U_1 and U_3 meets it at the height z_D with

$$W_{U_1}(y_1 - z_D) = W_{U_3}(y_3 - z_D). \qquad (A_D)$$

In a straight extended Śrī Yantra U_0 contains U_1, whose apex is the highest point y_{10} of the yantra, and D_0 contains D_1, so

$$z_D < y_0 < y_1 < \cdots < y_{10} < z_U; \tag{4}$$

equality is excluded by laminarity, since a tip on the apex of U_0 or D_0 is a point common to two legs of one family.

3.3 Laminarity as inequalities

Two triangles A, B of one family can have meeting legs only if their height ranges overlap. For downward triangles let A be the one with the lower apex and let $h = \min(b_A, b_B)$; for upward triangles let A be the one with the higher apex and $h = \max(b_A, b_B)$.

Lemma 3.2. *If the right legs of A and B do not meet, then $w_A(h) \geq w_B(h)$.*

Proof. Consider downward triangles; the upward case is the mirror image in a horizontal line. Since $a_A < a_B < h \leq \min(b_A, b_B)$, on the interval $a_B \leq y \leq h$ both right legs are present, and $w_A - w_B$ is an affine function of y. At $y = a_B$ it equals $w_A(a_B) > 0$. If $w_A(h) < w_B(h)$ the function vanishes in between, and the two legs meet. □

For the corners we use the containment in Definition 2.2. The half-width of U_0 at height y is the affine function $\ell_U(y)$ through (y_7, W_{D_1}) and (y_9, W_{D_3}), and for $y \leq y_7$,

$$W_T \leq \ell_U(y) \iff W_T\, y_{79} + W_{D_3}\,(y_7 - y) \leq W_{D_1}\,(y_9 - y), \tag{5}$$

with the analogous inequality for a corner of a downward triangle against ℓ_D.

Lemma 3.3. *In a straight extended Śrī Yantra the corner of every upward triangle $U_1, \ldots, U_4$ satisfies* (5) *at its base level, and the corner of every downward triangle $D_1, \ldots, D_5$ satisfies the analogous inequality for D_0.*

Proof. U_0 contains the upward triangles, so their corners lie in U_0; the upward bases lie at $y_1, \ldots, y_4 < y_7$. The downward case is the same. □

Proposition 3.4 (Reduction)**.** *Let Y be a straight extended Śrī Yantra of classical type, with levels $y_0, \dots, y_{10}$ and apexes z_D, z_U of D_0, U_0. Then the thirteen numbers satisfy the order* (4)*, the four quartic equations* (E_1)*–*(E_4)*, the collinearities* (C_D)*,* (C_U)*, the apex relations* (A_U)*,* (A_D)*, the sixteen inequalities of Lemma 3.2, one for each pair of triangles of one family among the nine, and the nine inequalities of Lemma 3.3, the half-widths being given by* (1)*.*

Proof. Lemma 3.1 gives (1) and (E_1)–(E_4). The collinearities and the apex relations restate the definition of the extension, and the order was established above. In each family every pair of the nine triangles has overlapping height ranges (Table 1), which gives the sixteen pairs of Lemma 3.2; laminarity provides its hypothesis. Lemma 3.3 gives the remaining nine. □

4 Thirteen points on the projective line

4.1 Weight

Put $p_0 = z_D$, $p_{i+1} = y_i$ for $i = 0, \dots, 10$, and $p_{12} = z_U$, and write $[a\,b] = |p_b - p_a|$, labelling the points by their levels $0, \dots, 10$ and by D and U. Every condition of Proposition 3.4 is, after substitution of (1) and division by one of its terms, of one of the four forms

$$m = 1, \qquad m_1 + m_2 = 1, \qquad m \geq 1, \qquad m_1 + m_2 \leq 1, \tag{6}$$

where m, m_1, m_2 are monomials $\prod [a\,b]^{e_{ab}}$ with integer exponents. The forms are listed in Tables 5 and 6 in the appendix. Call the *weight* of such a monomial at the point p_a the sum $\sum_b e_{ab}$ of the exponents of the brackets containing a.

Lemma 4.1. *Every monomial in Tables 5 and 6 has weight zero at each of the thirteen points. Consequently its value is unchanged when the thirteen points are moved by any real Möbius transformation μ with $\mu(p_a) \neq \infty$ for all a.*

Proof. The first statement is checked on the tables. For the second, $|\mu(p) - \mu(q)| = |\det \mu|\,|p - q| \,/\, (|\gamma p + \delta|\,|\gamma q + \delta|)$ for $\mu(x) = (\alpha x + \beta)/(\gamma x + \delta)$, and in a monomial of weight zero each factor $|\gamma p_a + \delta|$ occurs as often in the numerator as in the denominator, and so does $|\det \mu|$. □

The geometric reason is a symmetry of the straight figure: the projective transformations of the plane that fix the axis and the horizontal direction and commute with the reflection in the axis, $(x, y) \mapsto (\lambda x, ay + b)/(cy + d)$, carry straight extended Śrī Yantras to straight extended Śrī Yantras whenever the line $cy + d = 0$ misses the figure, and they act on the levels by Möbius transformations. Lemma 4.1 is the algebraic form of this statement, and it is all the proof uses.

4.2 The configuration space

Let $\mathcal{M}$ be the set of 13-tuples of distinct points of the real projective line in the cyclic order $p_0, p_1, \dots, p_{12}$, modulo the action of $\mathrm{PGL}_2(\mathbb{R})$. It is a cell of dimension ten, and by Lemma 4.1 each monomial of the tables is a well-defined function on it: a representative with all points finite exists, and a point at infinity may be admitted by omitting the brackets that contain it, which by weight zero occur equally often above and below. We define

$$\mathcal{V} = \{\text{the eight equations of Table 5 hold}\},$$
$$\mathcal{L} = \{\text{the twenty-five inequalities of Table 6 hold}\},$$

both subsets of $\mathcal{M}$. Eight equations in ten dimensions give $\mathcal{V}$ the expected dimension two.

Theorem 4.2. $\mathcal{V} \cap \mathcal{L} = \varnothing$.

Proof of Theorem A from Theorem 4.2. A straight extended Śrī Yantra of classical type gives thirteen points in the order (4), which lie in the cyclic order of $\mathcal{M}$, and by Proposition 3.4 and the tables the class of this configuration lies in $\mathcal{V} \cap \mathcal{L}$. □

Theorem 4.2 is stronger than Theorem A in two respects. The cyclic order of $\mathcal{M}$ is weaker than the linear order (4), and the inequalities of $\mathcal{L}$ are non-strict consequences of laminarity. Sections 5–7 prove it.

5 Cones

5.1 Gaps and Cartesian trees

Choose the representative in which $y_7 = \infty$. The other twelve points are then finite and in the linear order

$$P_0 < P_1 < \cdots < P_{11}, \qquad (P_0, \dots, P_{11}) = (y_8, y_9, y_{10}, z_U, z_D, y_0, y_1, \dots, y_6),$$

and the remaining freedom is the affine group of the line. Let $g_k = P_{k+1} - P_k > 0$ be the eleven gaps and $l_k = \log g_k$. Then $\mathcal{M}$ is identified with $\mathbb{R}^{11}$ modulo the diagonal translation, the translation acting on l as the scaling of the line. Every monomial of the tables involves only finite brackets in this representative, and each finite bracket is a sum of consecutive gaps,

$$[P_a\, P_b] = g_a + g_{a+1} + \cdots + g_{b-1} \qquad (a < b).$$

The *Cartesian tree* [5] of a sequence $l_0, \dots, l_{10}$ of distinct numbers is the binary tree on the indices whose root is the index of the largest entry and whose left and right subtrees are the Cartesian trees of the entries before and after it. For a binary tree T on $\{0, \dots, 10\}$ in symmetric order let

$$C_T = \{\, l : l_k \le l_{\pi(k)} \text{ for every non-root node } k \,\},$$

$\pi(k)$ denoting the parent of k. Every sequence lies in the cone of its Cartesian tree, ties lying on the common boundary of several cones, so the cones C_T cover $\mathcal{M}$. There are $C_{11} = 58{,}786$ binary trees on eleven nodes, the Catalan number, which is also the number of triangulations of a 13-gon. On C_T we normalise $l_{\text{root}} = 0$ and use the ten coordinates

$$s_k = l_k - l_{\pi(k)} \in (-\infty, 0], \qquad k \neq \text{root},$$

which identify C_T with the closed orthant $(-\infty, 0]^{10}$. The coordinates are independent, and every degeneration of the configuration, points merging or scales separating, is a direction in which some of the s_k tend to $-\infty$.

5.2 Distances on a cone

Lemma 5.1. *Let T be a binary tree, $a < b$, and let m be the lowest common ancestor in T of the nodes $a, \dots, b-1$. On C_T,*

$$\log[P_a\, P_b] = l_m + x_{ab}, \qquad x_{ab} = \log\Bigl(1 + \sum_{\substack{a \le k < b \\ k \neq m}} e^{\sigma_{mk}(s)}\Bigr), \tag{7}$$

where $\sigma_{mk}(s)$ is the sum of the coordinates s_j over the nodes $j \neq m$ on the path from m to k, and $l_m = \sigma_{\text{root},m}(s)$. In particular $0 \le x_{ab} \le \log(b-a)$, and every $\sigma_{mk} \le 0$.

Proof. In a Cartesian tree the largest entry of a contiguous range of indices sits at the lowest common ancestor of the range, so $l_m = \max_{a \le k < b} l_k$, and $l_k - l_m$ is the sum of the differences along the path, each of which is a coordinate $s_j \le 0$. Factor e^{l_m} out of the sum of the gaps. □

By (7), each logarithm of a monomial of the tables is, on C_T,

$$\log m = \lambda_m(s) + \sum_{a<b} e_{ab}\, x_{ab}(s),$$

an integer linear form in s plus a combination of the corrections x_{ab}. Each correction is a logarithm of a sum of exponentials of linear forms that are non-positive on the cone; it is monotone in every coordinate, takes its extreme values over a box at the corners of the box, and tends to an explicit limit as any set of coordinates tends to $-\infty$. In this sense each condition is *tropically linear* on each cone: a linear form, up to a bounded correction that vanishes when the configuration approaches the vertex of the cone at the rate at which the corresponding sums of gaps are dominated by their largest term.

Remark 5.2. The cones are charts around the vertices of the associahedron. The space $\mathcal{M}$ carries the 65 dihedral coordinates of Brown [1], one for each diagonal of a 13-gon whose sides are labelled by the thirteen points in cyclic order; its closure in these coordinates is the associahedron, a face is a set of pairwise non-crossing diagonals on which the corresponding coordinates vanish, and a vertex is a triangulation. The binary tree T and the triangulation corresponding to it describe the same degeneration, and C_T is a neighbourhood of that vertex in logarithmic coordinates. The gauge $y_7 = \infty$ serves only to make the coordinates explicit; every choice gives the same decomposition up to relabelling.

6 The vertex T^*

6.1 Where the equations and the inequalities meet

Let $\mathcal{V}_7 \supset \mathcal{V}$ be the set where the seven equations other than (E_1) hold. Its laminar part $\mathcal{V}_7 \cap \mathcal{L}$ is not empty; it is the family of figures such as Figure 2, laminar and satisfying everything except the one concurrence (t_2, t_6, t_6). In every computation we have made, the residual of (E_1) along this family is bounded away from zero except near one point of the boundary of $\mathcal{M}$, towards which it tends to zero. The point is the vertex T^* of the associahedron given by the triangulation of Figure 4: six nested clusters $F_k = \{z_D, y_0, \dots, y_k\}$, $k = 1, 2, 4, 6, 7, 8$, contract, the pairs $\{z_D, y_0\}$ and $\{y_{10}, z_U\}$ close up, which means that both tips approach the apexes of D_0 and U_0, and $\{y_3, y_4\}$ and $\{y_5, y_6\}$ close up as well. Along the deepest laminar figures we have found, all ten coordinates of T^* tend to zero together. On $\mathcal{V}_7 \cap \mathcal{L}$ the residual of (E_1) is therefore not bounded away from zero near T^*, and no subdivision in the coordinates of Section 5 can close a neighbourhood of T^* by bounding residuals; the neighbourhood is closed instead by an argument in coordinates adapted to the vertex.

6.2 The fan chart

Put $z_D = 0$, $z_U = \infty$ and $y_{10} = 1$, write $q_0 = z_D$ and $q_k = y_{k-1}$ for $k = 1, \dots, 11$, and set $t_k = q_k / q_{k+1} \in (0, 1)$ for $k = 1, \dots, 10$, so that $q_k = t_k t_{k+1} \cdots t_{10}$. Every bracket factors,

$$q_b - q_a = (t_b t_{b+1} \cdots t_{10})\, g(a, b), \qquad g(a, b) = 1 - t_a t_{a+1} \cdots t_{b-1} \quad (1 \le a < b),$$

and $q_b - q_0 = q_b$. Hence the logarithm of every monomial of the tables is $\sum_k \kappa_k \log t_k + \sum_{a<b} n_{ab} \log g(a, b)$ with integer κ and n. In this chart the six fan diagonals of T^* are the coordinates $t_2, t_3, t_5, t_7, t_8, t_9$, the pairs $\{z_D, y_0\}$ and $\{y_{10}, z_U\}$ correspond to $t_1 \to 0$ and $t_{10} \to 0$, and $\{y_3, y_4\}$, $\{y_5, y_6\}$ to $t_4 \to 1$ and $t_6 \to 1$.

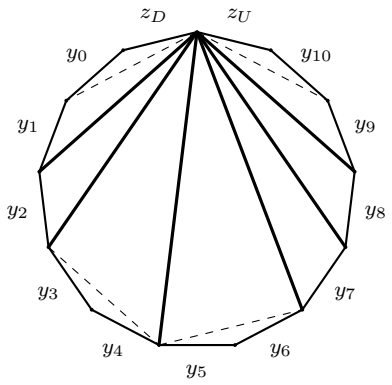


Figure 4: The vertex T^*. The sides of the 13-gon carry the thirteen points in their cyclic order, and a diagonal stands for the cluster of points on the sides it cuts off. Solid: the six fan diagonals $F_1, F_2, F_4, F_6, F_7, F_8$, $F_k = \{z_D, y_0, \dots, y_k\}$. Dashed: the diagonals of the pairs $\{z_D, y_0\}$, $\{y_{10}, z_U\}$, $\{y_3, y_4\}$ and $\{y_5, y_6\}$.

Let m_1, m_2 be the two monomials of (C_D) in Table 5, and let Φ be the combination

$$\Phi = \log m_{E_1} - \log m_{E_2} + \log m_{E_4} + \log(m_1 + m_2),$$

where m_{E_i} is the quotient of the two sides of (E_i) in Table 5. On $\mathcal{V}$ every term vanishes, so $\Phi = 0$ on $\mathcal{V}$.

Proposition 6.1. *In the fan chart, with $R = m_2/m_1$,*

$$\Phi = \log(1+R) + \sum_{(a,b)\in N^+} \log g(a,b) - \sum_{(a,b)\in N^-} \log g(a,b),$$

$$N^+ = \{(1,8),(2,4),(2,5),(3,7),(4,10),(5,8),(7,9),(8,11),(9,10)\},$$
$$N^- = \{(1,4),(2,8),(2,9),(3,5),(4,8),(5,7),(7,10),(8,10),(9,11)\};$$

the monomial part cancels. Where (A_U) holds, $R = t_9(1-t_8)/(1-t_9)$, so that $\log(1+R) = \log g(8,10) - \log g(9,10)$, and

$$\Phi = t_3t_4 + t_5t_6 + t_9t_{10} + \rho,$$

where every term of ρ has at least two factors among the fan coordinates $t_2, t_3, t_5, t_7, t_8, t_9$.

Proof. The first identity is the substitution of the factorisation into the four monomials; the exponents of $\log t_k$ cancel. Under (A_U) the two terms of $\log(1+R)$ cancel against $(8,10)$ and $(9,10)$, and $\Phi = \sum \pm \log(1-\pi_{ab})$ over the remaining eight pairs of each sign, with $\pi_{ab} = t_a \cdots t_{b-1}$. The three products $\pi_{35} = t_3t_4$, $\pi_{57} = t_5t_6$ and $\pi_{9,11} = t_9t_{10}$ in N^- contain one fan coordinate each; every other product contains at least two, and so does every term of order two or more in the expansion of $-\log(1-\pi)$. □

Near the face on which the six fan coordinates vanish, the equations (E_2)–(E_4), (A_U), (A_D) and (C_D) force, at leading order, $t_2 = t_8 = t_9 = t_3$, $t_5 = t_7 = t_3/t_6$ and $t_4 = t_6$; then $\Phi/t_3 \to t_4 + 1 + t_{10} \geq 1$, and $\Phi = 0$ is impossible. This is the mechanism by which $\mathcal{V}$ avoids the neighbourhood of T^* into which the laminar family runs. The certificate of the next subsection makes it rigorous on explicit regions.

6.3 The certified neighbourhood

In the adapted coordinates

$$e = \log t_3, \quad d = \log(t_6/t_3), \quad \log t_1, \quad \log t_{10},$$
$$r_2 = \log \frac{t_2}{t_3},\ r_4 = \log \frac{t_4}{t_6},\ r_5 = \log \frac{t_5 t_6}{t_3},\ r_7 = \log \frac{t_7 t_6}{t_3},\ r_8 = \log \frac{t_8}{t_3},\ r_9 = \log \frac{t_9}{t_3},$$

the linear parts of the six monomial equations do not involve the scale e, and each of them can be solved for one coordinate by interval propagation. The quantity Φ/t_3 is evaluated as a signed sum of terms $\pi\, h(\pi)/t_3$ and $R\,\tilde{h}(R)/t_3$ with $h(\pi) = -\log(1-\pi)/\pi$ and $\tilde{h}(R) = \log(1+R)/R$, which stays accurate as $t_3 \to 0$. A box of the adapted coordinates is refuted when propagation empties it or when one of the equations, one of the inequalities of $\mathcal{L}$, or Φ/t_3 has a sign on it.

Proposition 6.2. *$\mathcal{V} \cap \mathcal{L}$ has no point in the region $R = B_1 \cup B_2 \cup B_3 \cup D$ of the fan chart, where (t_1 and t_{10} arbitrary throughout)*

$$B_1 = \{t_3 \le 0.1,\ d \ge 2,\ |r_i| \le 2\}, \qquad B_2 = \{t_3 \le 0.1,\ 0.5 \le d \le 2.2,\ |r_i| \le 0.4\},$$
$$B_3 = \{0.1 \le t_3 \le 0.2,\ d \ge 1,\ |r_i| \le 0.25\}, \qquad D = \{t_3 \le 0.1,\ t_k \le 0.95 \text{ for } k = 2,5,7,8,9\},$$

with i ranging over $2,4,5,7,8,9$.

Certificate. Each of B_1, B_2, B_3 is covered by a branch and bound in the adapted coordinates in which every box is refuted, in 3,567, 81,685 and 1,485,235 boxes. The part of D outside $B_1 \cup B_2$ is covered by 25 pieces, $\{d \le 0.5\}$, the twelve pieces $\{0.5 \le d \le 2,\ \pm r_i \ge 0.4\}$ and the twelve pieces $\{d \ge 2,\ \pm r_i \ge 2\}$, each intersected with D, and every piece is closed in the same way, in 2,256,565 boxes in all; in D the bounds $t_k \le 0.95$ enter the propagation. The arithmetic is interval arithmetic with outward rounding on logarithms of positive quantities, sums being evaluated as log-sum-exp and differences only where their sign is known; it is described in Section 10. □

7 The certificate

7.1 Boxes

A *box* is a pair (T, β) of a binary tree and a product $\beta = \prod_k I_k$ of closed intervals $I_k \subset [-\infty, 0]$ in the coordinates s_k of C_T, an interval being allowed to be half-infinite, $I_k = [-\infty, c]$. A box is *closed* when one of the following tests shows that it contains no point of $\mathcal{V} \cap \mathcal{L}$; otherwise it is split.

(a) *Interval test.* By Lemma 5.1 the linear part of each condition is exact and each correction x_{ab} has an explicit enclosure on the box. If the enclosure of $\log m$ for an equation $m = 1$ excludes 0, or that of an inequality lies entirely on the wrong side, the box is closed.

(b) *Linear relaxation.* The variables are the ten coordinates s, bounded by the box, and the 66 corrections x_{ab}, bounded by their enclosures. For three points $i < j < k$, $L_{ik} = L_{ij} + \mathrm{sp}(L_{jk} - L_{ij})$ with $L = \log[\cdot\cdot]$ and $\mathrm{sp}(u) = \log(1 + e^u)$ convex; this is relaxed by tangent lines of sp from below and by its chord over the enclosure of $L_{jk} - L_{ij}$ from above (by $\mathrm{sp}(u) \le \mathrm{sp}(u_1)$ or $\mathrm{sp}(u) \le u + \mathrm{sp}(-u_0)$ when the enclosure is half-infinite). The six monomial equations are linear; the collinearities $m_1 + m_2 = 1$ and the corner clauses $m_1 + m_2 \le 1$ are relaxed by $m_1, m_2 \le 1$ and by tangent half-planes of the concave curve $e^u + e^v = 1$; the non-crossing clauses are linear. All slopes are dyadic rationals and every constant is computed for its rounded slope and rounded outwards, so each row is a valid

inequality on the whole box. The box is closed when the linear program is infeasible, as shown by Farkas multipliers $y \geq 0$, z with

$$w = A^{\mathsf{T}} y + E^{\mathsf{T}} z, \qquad \sum_j \min_{v_j \in [\underline{v}_j, \overline{v}_j]} w_j v_j \;>\; y^{\mathsf{T}} b + z^{\mathsf{T}} c,$$

the multipliers being computed in floating point [4] and the inequality checked in exact rational arithmetic.

(c) *Mean-value relaxation.* Each condition is a smooth function of the coordinates. A coordinate whose interval is half-infinite or wider than 1 is replaced inside the corrections by $u_k = e^{s_k} \in [e^{c_0}, e^{c_1}]$ (or $[0, e^{c_1}]$), in which the corrections are smooth up to $u_k = 0$. With the interval gradient G of the remaining smooth part h over the box, its centre z_c, half-widths r and a dyadic vector g_0,

$$h(z) - g_0 \cdot z \in \big[\, h(z_c) - g_0 \cdot z_c - R,\; h(z_c) - g_0 \cdot z_c + R \,\big], \qquad R = \sum_k \max |G_k - g_{0,k}|\, r_k,$$

which gives two linear rows for each equation and one for each inequality; trinomials whose two terms differ on the u-coordinates are written as $m_a + \mathrm{sp}(m_b - m_a)$ with an auxiliary variable for $m_b - m_a$ and the tangents and chord of sp. Infeasibility is certified as in (b).

(d) *Certified contraction.* When the system of (c) is feasible, each coordinate is minimised and maximised over it by linear programming, and a slab $\{s_k \leq \beta\}$ with β below the minimum is cut off the box if the system together with $s_k \leq \beta$ has an exact Farkas certificate; likewise at the upper end. Every point of $\mathcal{V} \cap \mathcal{L}$ in the box lies in the contracted box, with which the branch and bound continues.

(e) *Regions.* The fan coordinates of the box are enclosed, and the box is closed if the enclosure lies in the region R of Proposition 6.2.

A box that is not closed is split in one coordinate: after a failed test (c), in the coordinate with the largest share of the remainders R; otherwise in the coordinate whose correction terms are furthest from their chords. A finite interval is split where the range of e^{s_k} halves, a half-infinite one $[-\infty, c]$ at $c - \max(2, |c|)$.

Lemma 7.1. *If a box is closed by one of the tests (a)–(e), it contains no point of $\mathcal{V} \cap \mathcal{L}$.*

Proof. For (a) the enclosures contain the values of the conditions at every point of the box. For (b) and (c) every row is satisfied by every point of $\mathcal{V} \cap \mathcal{L}$ in the box, together with the values of the auxiliary variables at that point, and a Farkas certificate verified exactly shows that no point satisfies all rows. For (d) the same argument applies to each slab, and the slabs and the contracted box cover the box. For (e) it is Proposition 6.2. □

7.2 The run

Each of the 58,786 cones is first tested on its root box $(-\infty, 0]^{10}$: 9,159 are closed there by the interval test, every one of them by the corner clause of U_4. The remaining 49,627 are closed by branch and bound. In each of them the boxes form a binary tree whose leaves are the closed boxes, 2,285,836 in all; Table 2 summarises the run.

Proof of Theorem 4.2. The cones cover $\mathcal{M}$, the boxes of each cone cover it, and every box is closed. By Lemma 7.1 no box contains a point of $\mathcal{V} \cap \mathcal{L}$. □

cones	58,786
closed at the root	9,159
closed by branch and bound	49,627
boxes of the branch and bound	4,522,045
closed by the interval test (a)	154,707
closed by a linear certificate (b)	1,996,263
closed by the mean-value relaxation (c)	132,058
closed by the region D (e)	2,808
split	2,236,209
slabs cut off by certified contraction (d)	2,259,157
largest number of boxes in one cone	101,869

Table 2: The certificate of Theorem 4.2. Of the regions of Proposition 6.2 only D is used, in 16 cones.

Remark 7.2. The theorem was certified a second time by a different decomposition. There, a point at infinity is fixed and the space is cut into 32,805 cells by the ratios of four gaps between six of the twelve finite points; the other six are solved for in turn from equations linear in each, and boxes are closed by interval tests, by linear certificates of the kind in (b) over the 66 logarithmic distances, and by the regions of Proposition 6.2. Six cells, in which the four gap ratios are all extreme, resist this decomposition; they were closed by the cones of Section 5 restricted to the six cells (32,610 cones, 262,314 boxes, the region D used for 1,959 of them). Outside these six cells the two certifications share no decomposition.

II. Circular arcs

8 Circular-arc extended Śrī Yantras

8.1 The figure

We work with generalised circles $\gamma(x, y) = a(x^2 + y^2) + bx + cy + d$ as in [3, §3], so that lines are the case $a = 0$; the rim is the unit circle, and the axis is $x = 0$. A circular-arc Śrī Yantra is a realisation of the incidence structure of the nine triangles by arcs, symmetric about the axis [3, Definition 3.1]. In [3] its fourteen outer corners lie on the rim; here they are released, and the rim is instead met by the extension.

The extension of Section 2 takes the following form. The leg of U_0 is the generalised circle through the corners of D_1, D_2, D_3, and the leg of D_0 the one through the corners of U_1, U_2, U_3; their apexes are the poles $(0, 1)$ and $(0, -1)$ of the rim. The base of U_0 passes through the apex of D_1, and that of D_0 through the apex of U_1, as in the straight case. Between the yantra and the rim the large triangles then form a star, and its points are the cells of a new ring.

8.2 Walls

The new ring has two sites, one at each hand of the figure on the downward side and the upward side. On each site up to three further triangles can be carried out to the rim, namely the three outermost of the family, read down the page:

$$\text{down: } D_1,\ D_3,\ D_2, \qquad \text{up: } U_1,\ U_2,\ U_3.$$

A triangle is carried out to the rim by extending its legs until they meet a new base. Outermost means first in the order D_1, D_3, D_2 for the downward family, which reads down the page from the top of the figure, and in the order U_3, U_2, U_1 for the upward family, which reads up the page from the bottom. In each family the outermost carried-out triangle receives the axis-symmetric

arc through the tip of the figure, the apex of D_1 for the upward family and that of U_1 for the downward, and through the point where its own leg meets the rim; each further carried-out triangle, going inwards, receives the base given up by the one outside it; and U_0 (or D_0) receives the base given up by the innermost. Each carried-out triangle separates two cells of the ring, and we call it a *wall*. The figure is determined by its two sets of walls $W_D \subseteq \{D_1, D_3, D_2\}$ and $W_U \subseteq \{U_1, U_2, U_3\}$, and its ring has

$$2(|W_D| + 1) + 2(|W_U| + 1) + 2 = 2|W_D| + 2|W_U| + 6$$

cells, from 6 to 18. The outer node of each cell is an apex of U_0 or D_0 or a corner of a carried-out triangle, and it is required to lie on the rim.

Definition 8.1. A *circular-arc extended Śrī Yantra with walls* (W_D, W_U) is a circular-arc Śrī Yantra together with the extension and the carried-out triangles described above, in which the outer nodes of the new ring lie on the rim and the eleven triangles are laminar.

We encode a wall pattern by two octal digits du, the bits $1, 2, 4$ of d standing for D_1, D_3, D_2 and those of u for U_1, U_2, U_3. The two figures of Theorem B have the patterns 22 and 36.

9 Constructions

9.1 Moves

The constructions use the moves of [3, §4.2], here with the following names: placing a corner (`seed`) as a free point, as a height on the axis or as an angle on the rim; pinning a point at a pole (`pole`); the axis-symmetric generalised circle through two points (`s2`, move (B) of [3]); the generalised circle through three points (`s3`, move (C)); and the meeting of two generalised circles (`nx`, move (X)), of a generalised circle with the axis (`na`, move (A)) and with the rim (`nr`, move (R)). Where a meeting has two points the construction records which, as in [3]. An input may be the mirror image of an object already drawn, without that image being a separate step.

	`seed`	`pole`	`s2`	`s3`	`nx`	`na`	`nr`	steps
pattern 22	13	2	11	11	21	6	3	67
pattern 36	13	2	11	11	23	5	4	69

Table 3: The steps of the two constructions. Each draws 22 curves, the eleven bases and one leg of each mirror pair of legs, and 45, respectively 47, nodes; mirror images complete the figure.

9.2 The two constructions

Proof of Theorem B. The constructions are those of Tables 7 and 8 in Appendix B, the sequences `ARC_REC2` and `ARC_REC3` of the accompanying viewer; their moves are counted in Table 3 and their placed corners listed in Table 4. Of the 67 and 69 steps, 15 are placements, 13 corners and the two poles, which gives (i).

For (ii), each step draws one node or one curve from objects already drawn and realises the incidences listed with it: a curve drawn through two or three nodes passes through them, and a node drawn as a meeting lies on the curves it is the meeting of. Every incidence realised by a step therefore holds identically wherever the construction is defined, and the outer nodes of the ring lie on the rim because they are drawn there, either by a meeting with the rim or as the crossing of a leg with a base that was itself drawn through such a meeting. The steps realise 106 and 110 incidences of a node with a side, no incidence twice. That they are all the incidences of the figure is part of the computation for (iii).

pattern 22				pattern 36			
corner	placed		default	corner	placed		default
P_0	free	2	(0.1943, 0.0425)	P_0	free	2	(0.2863, 0.0617)
P_2	free	2	(0.0806, −0.0819)	P_1	free	2	(0.2153, −0.0301)
P_7	free	2	(0.0969, 0.0473)	P_3	axis	1	$y = -0.2609$
P_{20}	free	2	(0.2664, −0.0759)	P_{24}	free	2	(0.3989, 0.1417)
P_{27}	axis	1	$y = -0.3270$	P_{29}	free	2	(0.0989, 0.1659)
P_{29}	free	2	(0.0523, 0.1215)	P_{34}	free	2	(0.3181, −0.2469)
P_{36}	axis	1	$y = -0.5095$	P_{35}	free	2	(0.1596, −0.4869)
P_{48}	free	2	(0.4254, −0.1501)	P_{55}	axis	1	$y = -0.8863$
P_{53}	free	2	(0.0922, 0.2053)	P_{61}	axis	1	$y = 0.5139$
P_{56}	free	2	(0.1640, −0.5071)	P_{65}	free	2	(0.6663, −0.4304)
P_{65}	free	2	(0.5164, −0.3166)	P_{68}	axis	1	$y = 0.8706$
P_{68}	axis	1	$y = 0.7008$	ring corner 73	rim	1	$\theta = 0.7289$
ring corner 73	rim	1	$\theta = 0.3429$	ring corner 79	rim	1	$\theta = -0.7334$
total		22		total		20	

Table 4: The placed corners of the two constructions, the number of real parameters each carries, and its value at the figures of Figure 3. Node numbers are those of the accompanying files; the ring corners are numbered from 69, and a corner on the rim is given by its polar angle.

For (iii), the constructions were evaluated in interval arithmetic with outward rounding at sixty significant digits, from the corners of Table 4 read as exact decimals; a corner on the rim is the point $(\cos\theta, \sin\theta)$, enclosed by the Taylor series with its remainder. Every meeting the constructions take exists, its discriminant being enclosed in the positive reals, and the choice between its two points is decisive; every node is enclosed in a box of width below 10^{-33}. For every node and every side not joined by an incidence of the tables, the enclosure of the side's equation at the node excludes zero, so the figure has no incidence besides those realised by the steps. Each leg is the arc of its circle from its apex through the nodes on it to its corner. For each pair of a right leg and a right or left leg of another triangle of the same family, 50 pairs in each figure (the remaining pairs are their mirror images), either the two circles do not meet, the enclosure of the discriminant of their meeting lying below zero, or each of their two meeting points lies strictly on the far side of the chord of one of the two arcs from the nodes of that arc, and hence off that arc. This was the case for 15 and 35 pairs in pattern 22 and for 9 and 41 pairs in pattern 36. Hence no leg meets a leg of another triangle of its family, and the eleven triangles are laminar. □

Remark 9.1. The number of free parameters is the dimension of the family. Evaluating the incidence system of each figure, every node on every side it is required to lie on and every outer ring node on the rim, at the default figure, the Jacobian has rank equal to the number of unknowns less 22, respectively 20, with a gap of 10^{-1} against 2×10^{-15} in the singular values; the same computation returns 18 for the nine-triangle family of [3]. For pattern 22 the needs of the objects, counted as in [3, §7.1], sum to $128 = 22 + 106$, where 106 is the number of incidences: an order of the kind used here consumes each incidence exactly once, and none is arranged; for pattern 36 they sum to $130 = 20 + 110$.

10 Software and data

Everything the proofs use is in the repository of [3], `https://github.com/michaelkopp/sriyantra`: the viewer and the GeoGebra files at its top level, and the programs and data of this paper in the folder `extended/`.

The viewer and the GeoGebra files. The viewer `index.html` draws the two constructed figures of Theorem B from their placed corners, in the principal directions of the family or in the corners themselves, and exports them as SVG or PNG. The files `sri-yantra-arcs-extended-2.ggb` and `sri-yantra-arcs-extended-3.ggb` carry the two constructions as GeoGebra command lists, every step a `Circle` through points or an `Intersect` of two objects, evaluated when the file is opened. Re-evaluated from their command lists alone by an independent implementation of the operations, they reproduce all 69 nodes of the classical figure to 10^{-10}, every incidence to 1.7×10^{-15} and 4.4×10^{-16}, and the 10 and 14 outer nodes of the ring on the rim. Loaded into GeoGebra, they build 282 and 300 commands with nothing undefined, and after each of the 13 placed corners is dragged to 60% of its range every incidence still holds to 1.4×10^{-14}.

The cone certifier. In `extended/certifier/cones/`, `cone.py` implements the cones, the enclosures of Lemma 5.1, the interval test and the linear relaxation with its exact verification; `mvf.py` the mean-value relaxation and the certified contraction; `bbcone.py` the branch and bound on one cone; `roots_all.py` the test of the 58,786 root boxes; `runcones.py` and `run_all.sh` a resumable parallel run over all cones, and `summarize.py` the check that every cone has a closed record. The linear programs are solved by HiGHS [4] through SciPy; only their multipliers enter the certificate, and these are verified with Python's exact rationals. Eight test programs check soundness on samples: boxes, finite, half-infinite and wide, around Newton-polished points of $\mathcal{V}$ (with the inequalities left out) and of the laminar part of $\mathcal{V}_7$ (with (E_1) left out) must never be closed, the contracted box must contain the point, and the gradient enclosures of test (c) must contain the derivatives at random points; each test must print `PASS`, and `run_all.sh` runs five of them before it starts. The records of Table 2 amount to 71 hours of processor time on a desktop computer with ten cores; the records are in the same folder.

The fan-chart certifier. In `extended/certifier/fan/`, `fanRBB.js` and `fanD.js` certify the regions of Proposition 6.2, with the arithmetic of `lg.js`: every quantity is the logarithm of a positive number, enclosed in an interval with outward padding, sums are evaluated as log-sum-exp, and a difference is formed only where its sign is known, so that clusters of points at any scale are evaluated without loss.

The second certification. The cell decomposition of the remark in Section 7, with the records of its runs, is in `extended/certifier/cells/`.

Exact checks. In `extended/check/`, `straight.py` verifies Lemma 3.1 in exact rational arithmetic, `fan.py` the identities of Proposition 6.1, and `gentables.py` produces Tables 5 and 6 from the data the certifiers read, so that the tables printed here are the conditions certified.

The constructions of Theorem B. `extended/check/arcs/` holds the two constructions as read from the viewer (`recipes.json`), their evaluation in floating point and in interval arithmetic (`construct.py`, `iv.py`), the incidence lists `incidences_22.csv` and `incidences_36.csv`, the programs that write Table 4 (`seedtable.py`) and Tables 7 and 8 (`steptable.py`), and `certify.py`, which carries out the computation in the proof of Theorem B and writes `certify.log`. It reproduces the viewer's nodes to 3×10^{-14} at its default corners.

The figures. The figures of Part I are drawn by `extended/check/drawkites.py` and `drawstraight.py` from the points in the same folder, which `extended/check/fig1/` documents; those of Part II are exported from the viewer.

11 Concluding remarks

The straight extension fails, and it fails narrowly. Every condition but one can be met by laminar figures, and the one that remains, the concurrence of the leg of D_2 with the corner of U_4, is missed by an amount that becomes small only where the figure degenerates: near the vertex T^*, where both tips press against the apexes of the new triangles. There the miss is of first order in the size of the degeneration, and the combination Φ of Proposition 6.1 shows why it cannot vanish.

The method of Part I applies to any finite system of equations and inequalities between products of cross-ratios of points on a line. The space of such configurations is not compact, and the conditions degenerate at its boundary, but the cones of Section 5 cover it by finitely many orthants on each of which every condition is linear up to a correction that is bounded and explicit, and on which a branch and bound over half-infinite boxes terminates wherever the conditions are separated by a positive margin in these coordinates. What remains is local: the points of the boundary at which equations and inequalities meet, here the single vertex T^*, each of which requires its own analysis.

Four questions remain.

1. *A proof without a computer.* Numerically, the residual of (E_1) on the laminar part of $\mathcal{V}_7$ has infimum zero, approached only at T^*, so no polynomial certificate of a strict inequality can exist; a certificate has to vanish at T^* to the right order. Is there a short proof of Theorem 4.2 built around Proposition 6.1?

2. *The other patterns.* Theorem B gives constructions without arranged coincidences for two of the 64 wall patterns. Which of the others are realised by laminar figures, which of those admit such a construction, and do any require an arranged coincidence in the sense of [3, §7.1]?

3. *Rings on circles.* The ring of cells of an extended figure has its outer nodes on the rim. In [3] the grown rings of the nine-triangle figure were asked to have their tips on a circle, which costs a coincidence that has not been arranged by straightedge and compass. The same question can be asked of the rings that the extended figures grow outside the rim.

4. *Selection.* As for the nine-triangle figures, the choice of a member of each family is in the end aesthetic; the members drawn here were chosen by eye.

A The conditions in the thirteen points

Table 5: The eight equations in the thirteen points, $[a\,b] = |p_b - p_a|$, with a, b the levels $0, \dots, 10$ and $D = z_D$, $U = z_U$.

equation	form
E_1	$[1\,4][3\,7][4\,5][6\,8] = [1\,3][4\,6][4\,8][5\,7]$
E_2	$[0\,3][2\,4][2\,7][6\,9] = [0\,2][2\,6][3\,9][4\,7]$
E_3	$[0\,3][2\,7][4\,5][7\,9] = [0\,2][3\,9][4\,7][5\,7]$
E_4	$[1\,3][4\,8][5\,7][7\,10] = [1\,7][3\,7][4\,5][8\,10]$
A_U	$[0\,7][2\,7][4\,5][9\,U] = [0\,2][4\,9][5\,7][7\,U]$
A_D	$[D\,1][3\,10][4\,8][5\,7] = [D\,3][1\,7][4\,5][8\,10]$
C_U	$1 = \dfrac{[0\,2][1\,7][2\,3][3\,9]}{[0\,3][1\,3][2\,7][2\,9]} + \dfrac{[0\,2][1\,2][3\,9][3\,10][4\,8][5\,7]}{[0\,3][1\,3][2\,7][2\,9][4\,5][8\,10]}$
C_D	$1 = \dfrac{[0\,7][1\,3][2\,7][8\,9]}{[0\,2][1\,8][3\,7][7\,9]} + \dfrac{[1\,3][4\,9][5\,7][7\,8]}{[1\,8][3\,7][4\,5][7\,9]}$

Table 6: The twenty-five inequalities that define $\mathcal{L}$. A non-crossing clause nc AB reads $m \geq 1$; a corner clause reads $m_1 + m_2 \leq 1$.

clause	form
non-crossing D_1D_2	$[0\,7][1\,3][2\,7] \geq [0\,2][1\,7][3\,7]$
non-crossing D_1D_3	$[0\,7][2\,7][4\,5] \geq [0\,2][4\,7][5\,7]$
non-crossing D_1D_4	$[0\,6][2\,4][2\,7] \geq [0\,2][2\,6][4\,7]$
non-crossing D_1D_5	$[0\,5][2\,7][4\,5][6\,8] \geq [0\,2][4\,6][5\,7][5\,8]$
non-crossing D_2D_3	$[1\,8][3\,7][4\,5] \geq [1\,3][4\,8][5\,7]$
non-crossing D_2D_4	$[1\,6][2\,4][3\,7] \geq [1\,3][2\,6][4\,7]$
non-crossing D_2D_5	$[1\,5][3\,7][4\,5][6\,8] \geq [1\,3][4\,6][5\,7][5\,8]$
non-crossing D_4D_3	$[2\,6][4\,5][4\,7] \geq [2\,4][4\,6][5\,7]$
non-crossing D_4D_5	$[2\,5][4\,5][4\,7][6\,8] \geq [2\,4][4\,6][5\,7][5\,8]$
non-crossing D_5D_3	$[4\,6][5\,8] \geq [4\,5][6\,8]$
non-crossing U_1U_2	$[0\,2][3\,10][4\,8][5\,7] \geq [0\,3][2\,7][4\,5][8\,10]$
non-crossing U_1U_3	$[3\,10][4\,8][5\,7] \geq [3\,7][4\,5][8\,10]$
non-crossing U_1U_4	$[4\,10][6\,8] \geq [4\,6][8\,10]$
non-crossing U_2U_3	$[0\,3][2\,9] \geq [0\,2][3\,9]$
non-crossing U_2U_4	$[0\,3][2\,7][4\,5][4\,9][6\,8] \geq [0\,2][3\,9][4\,6][4\,8][5\,7]$
non-crossing U_4U_3	$[4\,6][4\,8][5\,7] \geq [4\,5][4\,7][6\,8]$
corner U_1	$\frac{[0\,2][3\,10][4\,8][5\,7][7\,9]}{[0\,7][2\,7][3\,9][4\,5][8\,10]} + \frac{[0\,2][3\,7][4\,9][5\,7]}{[0\,7][2\,7][3\,9][4\,5]} \leq 1$
corner U_2	$\frac{[0\,3][7\,9]}{[0\,7][3\,9]} + \frac{[0\,2][4\,9][5\,7]}{[0\,7][2\,9][4\,5]} \leq 1$
corner U_3	$\frac{[0\,2][1\,7][7\,9]}{[0\,7][1\,9][2\,7]} + \frac{[0\,2][1\,7][4\,9][5\,7]}{[0\,7][1\,9][2\,7][4\,5]} \leq 1$
corner U_4	$\frac{[0\,2][4\,6][4\,8][5\,7][7\,9]}{[0\,7][2\,7][4\,5][4\,9][6\,8]} + \frac{[0\,2][4\,7][5\,7]}{[0\,7][2\,7][4\,5]} \leq 1$
corner D_1	$\frac{[0\,7][1\,3][2\,7][4\,5][8\,10]}{[0\,2][1\,7][3\,10][4\,8][5\,7]} + \frac{[3\,7][4\,5][8\,10]}{[3\,10][4\,8][5\,7]} \leq 1$
corner D_2	$\frac{[3\,7][4\,5][8\,10]}{[3\,10][4\,8][5\,7]} + \frac{[1\,7][3\,8][4\,5][8\,10]}{[1\,8][3\,10][4\,8][5\,7]} \leq 1$
corner D_3	$\frac{[1\,3][4\,9][8\,10]}{[1\,9][3\,10][4\,8]} + \frac{[1\,7][3\,9][4\,5][8\,10]}{[1\,9][3\,10][4\,8][5\,7]} \leq 1$
corner D_4	$\frac{[1\,3][2\,6][4\,5][4\,7][8\,10]}{[1\,6][2\,4][3\,10][4\,8][5\,7]} + \frac{[1\,7][3\,6][4\,5][8\,10]}{[1\,6][3\,10][4\,8][5\,7]} \leq 1$
corner D_5	$\frac{[1\,3][4\,6][5\,8][8\,10]}{[1\,5][3\,10][4\,8][6\,8]} + \frac{[1\,7][3\,5][4\,5][8\,10]}{[1\,5][3\,10][4\,8][5\,7]} \leq 1$

B The two constructions

Tables 7 and 8 list the steps of the constructions of Theorem B in the order in which they are taken. Each row gives the object drawn and the objects it is drawn from; the incidences a step realises are those between the object drawn and the objects listed with it.

	drawn	from	position		drawn	from	position
1	P_{69}	pole	$(0.000, -1.000)$	35	$\beta(D_0)$	P_{61}, P_{73}	
2	P_{70}	pole	$(0.000, 1.000)$	36	P_{31}	$\beta(D_0) \cap \lambda(U_0)$	$(0.511, 0.344)$
3	P_{65}	placed	$(0.516, -0.317)$	37	$\lambda(D_3)$	P_{20}, P_{31}, P_{36}	
4	P_{73}	placed on the rim	$(0.942, 0.336)$	38	P_{71}	$\lambda(D_3) \cap \text{rim}$	$(0.711, 0.703)$
5	$\lambda(D_0)$	P_{65}, P_{69}, P_{73}		39	$\beta(D_3)$	P_{68}, P_{71}	
6	P_{56}	placed	$(0.164, -0.507)$	40	P_{35}	$\beta(U_0) \cap \lambda(D_3)$	$(0.114, -0.326)$
7	P_{29}	placed	$(0.052, 0.121)$	41	P_{34}	$\lambda(D_3) \cap \lambda(U_3)$	$(0.215, -0.161)$
8	P_{20}	placed	$(0.266, -0.076)$	42	P_{32}	$\beta(D_2) \cap \lambda(D_3)$	$(0.423, 0.191)$
9	P_{48}	placed	$(0.425, -0.150)$	43	$\beta(U_1)$	P_{34}, P_{48}	
10	P_{53}	placed	$(0.092, 0.205)$	44	P_{47}	$\beta(U_1) \cap \lambda(D_0)$	$(0.643, -0.132)$
11	P_2	placed	$(0.081, -0.082)$	45	P_3	$\beta(U_1) \cap \text{axis}$	$(0.000, -0.164)$
12	$\beta(U_4)$	P_2, P_{20}		46	$\lambda(U_1)$	P_{32}, P_{47}, P_{68}	
13	P_{16}	$\beta(U_4) \cap \text{axis}$	$(0.000, -0.083)$	47	$\lambda(D_5)$	P_0, P_2, P_3	
14	P_{36}	placed on the axis	$(0.000, -0.509)$	48	P_{13}	$\beta(D_0) \cap \lambda(U_1)$	$(0.305, 0.347)$
15	$\beta(U_3)$	P_{36}, P_{56}		49	P_1	$\lambda(D_5) \cap \lambda(U_3)$	$(0.137, -0.021)$
16	P_{44}	$\beta(U_3) \cap \lambda(D_0)$	$(0.388, -0.496)$	50	P_{57}	$\lambda(D_2) \cap \lambda(U_1)$	$(0.542, 0.023)$
17	P_{27}	placed on the axis	$(0.000, -0.327)$	51	$\lambda(D_1)$	P_7, P_{13}, P_{16}	
18	$\beta(U_0)$	P_{27}, P_{65}		52	P_9	$\lambda(D_1) \cap \lambda(U_0)$	$(0.400, 0.493)$
19	P_{75}	$\beta(U_0) \cap \text{rim}$	$(0.957, -0.291)$	53	P_{15}	$\lambda(D_1) \cap \lambda(U_4)$	$(0.148, 0.118)$
20	P_7	placed	$(0.097, 0.047)$	54	$\beta(D_4)$	P_{15}, P_{29}	
21	$\lambda(U_3)$	P_7, P_{29}, P_{44}		55	P_{14}	$\beta(D_2) \cap \lambda(D_1)$	$(0.207, 0.202)$
22	P_{46}	$\lambda(U_3) \cap \text{axis}$	$(0.000, 0.206)$	56	$\lambda(U_2)$	P_{14}, P_{48}, P_{65}	
23	$\beta(D_2)$	P_{46}, P_{53}		57	P_{67}	$\lambda(U_2) \cap \text{axis}$	$(0.000, 0.491)$
24	P_{21}	$\beta(U_4) \cap \lambda(U_3)$	$(0.171, -0.080)$	58	P_{33}	$\lambda(D_3) \cap \lambda(U_2)$	$(0.324, 0.021)$
25	P_{45}	$\beta(U_0) \cap \lambda(U_3)$	$(0.302, -0.323)$	59	P_{38}	$\beta(D_0) \cap \lambda(U_2)$	$(0.107, 0.348)$
26	$\lambda(D_2)$	P_{45}, P_{48}, P_{56}		60	$\beta(D_1)$	P_9, P_{67}	
27	P_{55}	$\lambda(D_2) \cap \text{axis}$	$(0.000, -0.712)$	61	P_{10}	$\beta(D_1) \cap \lambda(U_1)$	$(0.186, 0.492)$
28	P_{52}	$\beta(D_2) \cap \lambda(D_2)$	$(0.637, 0.172)$	62	P_{77}	$\lambda(U_2) \cap \text{rim}$	$(0.704, -0.710)$
29	$\lambda(U_0)$	P_{52}, P_{70}, P_{75}		63	$\beta(U_2)$	P_{55}, P_{77}	
30	P_{68}	placed on the axis	$(0.000, 0.701)$	64	P_{24}	$\beta(D_4) \cap \lambda(U_2)$	$(0.270, 0.107)$
31	P_0	placed	$(0.194, 0.043)$	65	$\lambda(D_4)$	P_{21}, P_{24}, P_{27}	
32	$\beta(D_5)$	P_0, P_7		66	P_{26}	$\beta(U_1) \cap \lambda(D_4)$	$(0.118, -0.163)$
33	$\lambda(U_4)$	P_0, P_{20}, P_{53}		67	P_{25}	$\lambda(D_4) \cap \lambda(U_4)$	$(0.218, 0.004)$
34	P_{61}	$\lambda(U_4) \cap \text{axis}$	$(0.000, 0.348)$				

Table 7: The construction of pattern 22, step by step. $\beta(T)$ is the base of T, $\lambda(T)$ its right leg and $\lambda'(T)$ its left leg, and P' is the mirror image of P. A curve is drawn through the nodes listed, and a node as the meeting listed; of the two points of a meeting it is the one at the position given, which is its position at the corners of Table 4 to three decimals.

	drawn	from	position		drawn	from	position
1	P_{69}	pole	$(0.000, -1.000)$	36	P_{67}	$\lambda(U_2) \cap$ axis	$(0.000, 0.693)$
2	P_{70}	pole	$(0.000, 1.000)$	37	$\beta(D_3)$	P_{67}, P_{73}	
3	P_{73}	placed on the rim	$(0.746, 0.666)$	38	$\lambda(D_0)$	P_{44}, P_{65}, P_{69}	
4	P_{68}	placed on the axis	$(0.000, 0.871)$	39	P_{47}	$\beta(U_1) \cap \lambda(D_0)$	$(0.803, -0.171)$
5	P_{61}	placed on the axis	$(0.000, 0.514)$	40	P_{75}	$\lambda(D_0) \cap$ rim	$(0.920, 0.392)$
6	P_{35}	placed	$(0.160, -0.487)$	41	$\beta(D_0)$	P_{61}, P_{75}	
7	P_{24}	placed	$(0.399, 0.142)$	42	P_{31}	$\beta(D_0) \cap \lambda(D_3)$	$(0.667, 0.450)$
8	P_3	placed on the axis	$(0.000, -0.261)$	43	P_{38}	$\beta(D_0) \cap \lambda(U_2)$	$(0.151, 0.511)$
9	P_0	placed	$(0.286, 0.062)$	44	$\lambda(U_0)$	P_{31}, P_{70}, P_{77}	
10	P_{29}	placed	$(0.099, 0.166)$	45	P_9	$\beta(D_3) \cap \lambda(U_0)$	$(0.474, 0.682)$
11	$\beta(D_4)$	P_{24}, P_{29}		46	P_{52}	$\lambda(D_2) \cap \lambda(U_0)$	$(0.803, 0.202)$
12	P_{34}	placed	$(0.318, -0.247)$	47	$\beta(D_2)$	P_{46}, P_{52}	
13	$\lambda(D_3)$	P_{34}, P_{35}, P_{73}		48	P_{14}	$\beta(D_2) \cap \lambda(U_2)$	$(0.307, 0.291)$
14	$\beta(U_1)$	P_3, P_{34}		49	P_{32}	$\beta(D_2) \cap \lambda(D_3)$	$(0.583, 0.252)$
15	P_{36}	$\lambda(D_3) \cap$ axis	$(0.000, -0.699)$	50	$\lambda(U_1)$	P_{32}, P_{47}, P_{68}	
16	P_1	placed	$(0.215, -0.030)$	51	P_{57}	$\lambda(D_2) \cap \lambda(U_1)$	$(0.716, 0.021)$
17	$\lambda(U_3)$	P_1, P_{29}, P_{34}		52	P_{10}	$\beta(D_3) \cap \lambda(U_1)$	$(0.215, 0.691)$
18	$\lambda(D_5)$	P_0, P_1, P_3		53	P_{13}	$\beta(D_0) \cap \lambda(U_1)$	$(0.405, 0.491)$
19	P_{81}	$\lambda(U_3) \cap$ rim	$(0.488, -0.873)$	54	$\lambda(D_1)$	P_9, P_{13}, P_{14}	
20	P_{46}	$\lambda(U_3) \cap$ axis	$(0.000, 0.306)$	55	P_7	$\lambda(D_1) \cap \lambda(U_3)$	$(0.160, 0.067)$
21	P_{65}	placed	$(0.666, -0.430)$	56	P_{15}	$\beta(D_4) \cap \lambda(D_1)$	$(0.226, 0.159)$
22	$\beta(U_0)$	P_{35}, P_{65}		57	P_{71}	$\lambda(D_1) \cap$ rim	$(0.519, 0.855)$
23	P_{45}	$\beta(U_0) \cap \lambda(U_3)$	$(0.399, -0.469)$	58	$\beta(D_1)$	P_{68}, P_{71}	
24	P_{77}	$\beta(U_0) \cap$ rim	$(0.928, -0.373)$	59	$\beta(D_5)$	P_0, P_7	
25	P_{27}	$\beta(U_0) \cap$ axis	$(0.000, -0.490)$	60	P_{16}	$\lambda(D_1) \cap$ axis	$(0.000, -0.120)$
26	P_{55}	placed on the axis	$(0.000, -0.886)$	61	$\lambda(U_4)$	P_0, P_{15}, P_{61}	
27	$\beta(U_3)$	P_{55}, P_{81}		62	P_{53}	$\beta(D_2) \cap \lambda(U_4)$	$(0.135, 0.303)$
28	P_{79}	placed on the rim	$(0.743, -0.669)$	63	P_{20}	$\lambda(D_3) \cap \lambda(U_4)$	$(0.395, -0.117)$
29	$\lambda(U_2)$	P_{24}, P_{65}, P_{79}		64	$\beta(U_4)$	P_{16}, P_{20}	
30	$\beta(U_2)$	P_{36}, P_{79}		65	P_2	$\beta(U_4) \cap \lambda(D_5)$	$(0.139, -0.119)$
31	P_{33}	$\lambda(D_3) \cap \lambda(U_2)$	$(0.468, 0.017)$	66	P_{21}	$\beta(U_4) \cap \lambda(U_3)$	$(0.260, -0.118)$
32	P_{48}	$\beta(U_1) \cap \lambda(U_2)$	$(0.580, -0.214)$	67	$\lambda(D_4)$	P_{21}, P_{24}, P_{27}	
33	P_{44}	$\beta(U_2) \cap \lambda(U_3)$	$(0.456, -0.688)$	68	P_{26}	$\beta(U_1) \cap \lambda(D_4)$	$(0.173, -0.257)$
34	$\lambda(D_2)$	P_{45}, P_{48}, P_{55}		69	P_{25}	$\lambda(D_4) \cap \lambda(U_4)$	$(0.326, -0.003)$
35	P_{56}	$\beta(U_2) \cap \lambda(D_2)$	$(0.200, -0.697)$				

Table 8: The construction of pattern 36, step by step. $\beta(T)$ is the base of T, $\lambda(T)$ its right leg and $\lambda'(T)$ its left leg, and P' is the mirror image of P. A curve is drawn through the nodes listed, and a node as the meeting listed; of the two points of a meeting it is the one at the position given, which is its position at the corners of Table 4 to three decimals.

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